ROOFSDK 4.0.4

Build and Installation Manual

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Approved

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1. SCOPE

This document describes how to build and install the ROOFSDK.

2. APPLICABLE DOCUMENTS

Ref.	Acronym	Title	Originator	Id/Rev

Table 1: Applicable documents

3. BUILD ROOFSDK FROM SOURCE

In order to build ROOFSDK, define the following environment variables:

- **DEV_ROOT** = <Path where roofsdk directory is located>
  (Ex. ${HOME}/roofsdk)
- **RBM_VER** = <Version of the roof build machine> (Ex. 4.0.2)
- **ROOFSDK_DEST_DIR**= <path where you want the SDK to be installed>
- **CMAKE_ADDITIONAL_MODULES_DIR** = $DEV_ROOT/cmake_modules
- **QT_SELECT** = 5
- **QTDIR** = <Path where qt directory is installed>
  (Ex ${HOME}/Qt5.15.2/5.15.2/gcc_64
- **Qt5_DIR** = $QTDIR/lib/cmake/Qt5

Execute the following commands:

- cd ${DEV_ROOT}/roofsdk
- mkdir build
- cd build
- cmake ../ -DCMAKE_INSTALL_PREFIX=${ROOFSDK_DEST_DIR}
- make
- make install

4. ROOFSDK INSTALLATION

4.1. WINDOWS 10

Supported compiler is Visual Studio 2019.

4.1.1. Pre-requirements

Install last version of CMake.
It can be found at: (*https://cmake.org/download/)*

Install last version of Code Meter Runtime.
It can be found at (*https://www.wibu.com/it/support/user/downloads-user-software.html)*

4.1.2. Installation

Two packages will be provided:

- roofsdk package (roofsdk_4.0.4_xxx.zip)
- dependencies package (WRBM_4.0.4_stack.zip)

Extract the dependencies package in C:/<DIRDEP>
Define the following environment variables:

- **ROOF3RDPARTY** = C:/<DIRDEP>
- **BOOST_ROOT** = C:/<DIRDEP>/boost_1_72_0
- **BOOST_LIBRARYDIR** = C:/<DIRDEP>/boost_1_72_0/lib64-msvc-14.2
- **OPENSSL_ROOT_DIR** = C:/<DIRDEP>/OpenSSL-Win64
- **POCO_DIR** = C:/<DIRDEP>/Poco
- **QT_PLUGIN_PATH** = C:/<DIRDEP>/Qt5.12.7/5.12.7/msvc2019_64/plugins
- **Qt5_DIR** = C:/<DIRDEP>/Qt5.12.7/5.12.7/msvc2019_64/lib/cmake/Qt5
- **QTDIR** = C:/<DIRDEP>/Qt5.12.7/5.12.7/msvc2019_64
- **ZMQ_ROOT** = C:/<DIRDEP>/zmq432

Extract the roofsdk package to a directory of your choice C:/<DIRROOF>
Define the following environment variables:

- **ROOFSDK_DIR** = C:/<DIRROOF>/roofsdk_4.0.4

Add to **Path** environment variable the following paths:

- C:\<DIRROOF>\roofsdk_4.0.4\lib
- C:\<DIRDEP>\boost_1_72_0\lib64-msvc-14.2
- C:\<DIRDEP>\Poco\bin
- C:\<DIRDEP>\zmq\lib

- C:\<DIRDEP>\fftw3
- C:\<DIRDEP>\OpenSSL-Win64
- C:\<DIRDEP>\Qt5.12.7/5.12.7/msvc2019_64/bin
- C:\<DIRDEP>\WIBU

4.1.3. Examples

Examples of ROOFSDK usage can be found in the "examples" folder. To compile the generic example, use CMake as building system and VisualStudio2019

4.2. UBUNTU

4.2.1. Pre-requirements

The minimum required dependencies to use the ROOFSDK core module (roofcore) are:

- Cmake (3.22 or higher)
- Boost (1.71.0)
- Poco (1.11.1)
- CodeMeter Runtime (7.20 or higher)

The in order to use additional modules of ROOFSDK check the table below:

Item	***Module***	***Libraries***	***Additional dependencies***
1	ROOFQCore	roofqcore	• Zmq (4.3.5) • Qt5 (5.15.2) • OpenSSL (Reinstall Poco to install crypto module)
2	ROOF IPC	roofipc	• ZMQ (4.3.5)
3	SMS Messages	roofifmsg emsscrtlmsg	• ZMQ (4.3.5)
4	QSMS Messages	emssqcrtlmsg	• ZMQ (4.3.5) • Qt5

Table 1: Additional modules dependecies

In case QT Libraries are needed, install them in =${HOME}/Qt5.15.2
After installing the ZMQ Library, install the ZMQ CPP wrapper or copy the zmq.hpp file in /usr/include.

4.2.2. Installation

Extract the ROOFSDK package (roofsdk_4.0.4_xxx.tgz) to a directory of your choice "<DIR>".
Define the following environment variables:

- **ROOFSDK_DIR** = <DIR>/roofsdk_4.0.4
- **ROOFSDK_BIN_DIR** = ${ROOFSDK_DIR}/bin
- **ROOFSDK_LIB_DIR** = ${ROOFSDK_DIR}/lib
- **ROOFSDK_INCLUDE_DIR** = ${ROOFSDK_DIR}/include
- **QTDIR=**${HOME}/Qt5.15.2/5.15.2/gcc_64
- **Qt5_DIR=**${QTDIR}/lib/cmake/Qt5
- **QT_LIB_DIR=**${QTDIR}/lib

Add in the **LD_LIBRARY_PATH** the following variables:

- **ROOFSDK_LIB_DIR**
- **QT_LIB_DIR**

4.2.3. Examples

Examples of ROOFSDK usage can be found in the "examples" folder. They use CMake as a building system. To compile the generic example, just open a terminal and go to the folder for the example you want to compile and implement the following steps:

- mkdir build
- cd build
- cmake ../
- make

4.3. RED HAT

4.3.1. Pre-requirements

The minimum required dependencies to use the ROOFSDK core module (roofcore) are:

- Cmake (3.22 or higher)
- Boost (1.66.0)
- Poco (1.11.2)
- CodeMeter Runtime (7.20 or higher)
- CodeMeter AxProtector (10.70 or higher)
- Codemeter AxProtector-dev (10.70 or higher)

The in order to use additional modules of ROOFSDK check the table below:

Item	***Module***	***Libraries***	***Additional dependencies***
1	ROOFQCore	roofqcore	• Z (4.3.4) • Qt5 (5.15.2) • OpenSSL (Reinstall Poco to install crypto module)
2	ROOF IPC	roofipc	• ZMQ (4.3.4)
3	SMS Messages	roofifmsg emsscrtlmsg	• ZMQ (4.3.4)
4	QSMS Messages	emssqcrtlmsg	• ZMQ (4.3.4) • Qt5

Table 2: Additional modules dependecies

After installing the ZMQ Library, install the ZMQ CPP wrapper or copy the zmq.hpp file in /usr/include.

4.3.2. Installation

Extract the ROOFSDK package (roofsdk_4.0.4_xxx.tgz) to a directory of your choice "<DIR>".
Define the following environment variables:

- **ROOFSDK_DIR** = <DIR>/roofsdk_4.0.4
- **ROOFSDK_BIN_DIR** = ${ROOFSDK_DIR}/bin
- **ROOFSDK_LIB_DIR** = ${ROOFSDK_DIR}/lib
- **ROOFSDK_INCLUDE_DIR** = ${ROOFSDK_DIR}/include

Add in the **LD_LIBRARY_PATH** the following variables:

- **ROOFSDK_LIB_DIR**

4.3.3. Examples

Examples of ROOFSDK usage can be found in the "examples" folder. They use CMake as a building system. To compile the generic example, just open a terminal and go to the folder for the example you want to compile and implement the following steps:

- mkdir build
- cd build
- cmake ../
- make

5. ROOFSDK LICENSE LOG

The licenses are checked at start-up and while the application is running. The license status is reported in the *roofsdklicense.log* file, which is created in the same path of the application binary. The log file shows the product code and the expiration date of each license. Below is an example of the *roofsdklicense.log* file

```
13-01-2021 13:23:16 INFO  License for Product code = 3001 found.
13-01-2021 13:23:16 INFO  Product code = 3000 Expiration time = 20211231T225959
13-01-2021 13:23:16 INFO  License for Product code = 3002 found.
13-01-2021 13:23:16 INFO  Product code = 3001 Expiration time = 20220111T094735
13-01-2021 13:23:16 INFO  License for Product code = 3004 found.
13-01-2021 13:23:16 INFO  Product code = 3003 Expiration time = 20220111T094910
13-01-2021 13:23:18 INFO  Run thread
13-01-2021 13:23:18 Product code = 3000 Expiration time(UTC) = 20211231T225959
13-01-2021 13:23:18 Product code = 3001 Expiration time(UTC) = 20220111T094735
13-01-2021 13:23:18 Product code = 3003 Expiration time(UTC) = 20220111T094910
```

If a license is missing for any of the roofsdk libraries, the error is reported in the *roofsdklicense.log* file as shown below

```
13-01-2021 14:02:48 INFO  License for Product code = 3001 found.
13-01-2021 14:02:48 INFO  Product code = 3000 Expiration time = 20211231T225959
13-01-2021 14:02:48 INFO  License for Product code = 3002 found.
13-01-2021 14:02:49 INFO  Product code = 3001 Expiration time = 20220111T094735
13-01-2021 14:02:49 ERROR  License for Product code = 3004 not found.
```

In a Windows environment, the OS will also notify the error with the popup shown in Figure 1

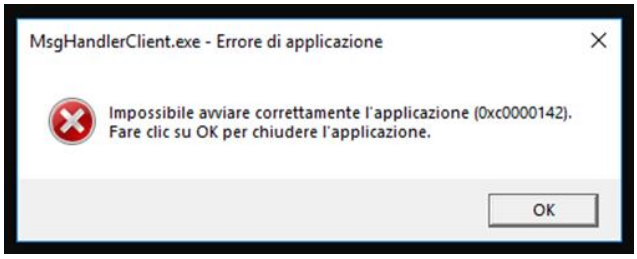

Figure 1 – Windows OS error popup due to license unavailability

Further information can be retrieved from the CodeMeter Control Center Event tab, as shown in Figure 2

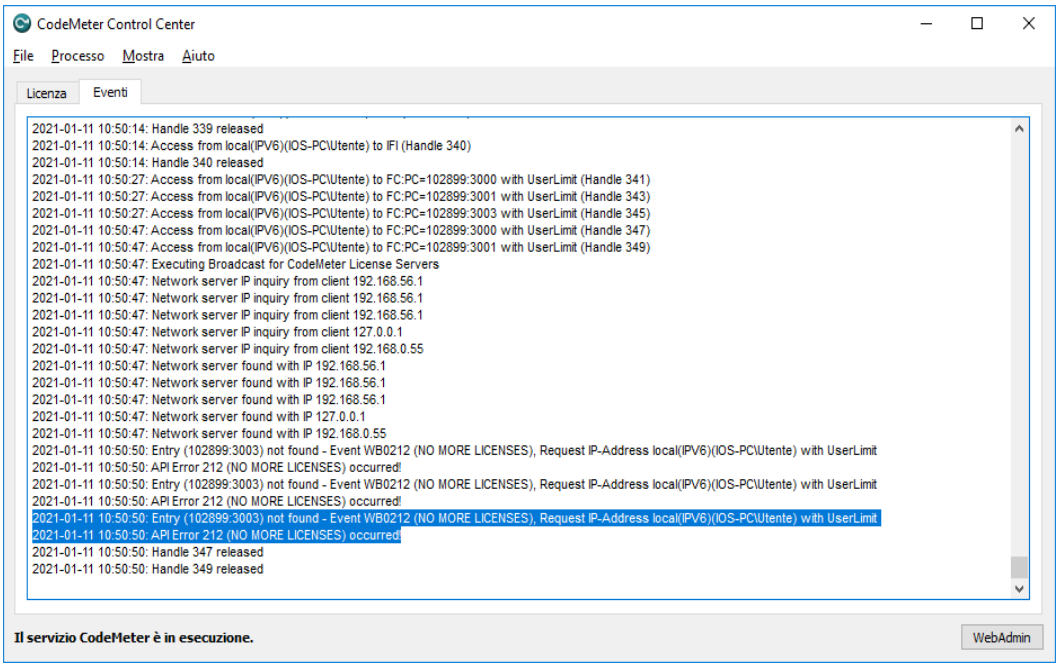

Figure 2 – CodeMeter Control Center Event tab